

9.1 ALGEBRAIC EXPRESSIONS

Define an algebraic equation

An algebraic sentence involving the sign of equality “=” is called an algebraic equation or simply an equation e.g.

- (i) $6 - 2 = 4$
- (ii) $2x + 4 = 8$

are examples of an algebraic equation.

An equation can be compared to the pans of common balance in equilibrium. The two sides of the equation are like the two scale pans and the equality sign shows that scale pans are balanced.



In equation ' $2x + 4 = 8$ ' the left hand side $2x + 4$ is equal to the right hand side 8.

Differentiate between equation and an expression

An algebraic expression is a combination of numbers, variable and arithmetical operations.

Thus $2a$, y , $31 + z$, $2x - 3y \times 4$ are all examples of algebraic expressions. Whereas two expressions connected by sign of equality is called an equation. In an equation, the two expressions are equated with the symbol “=”.

$2x + 3 = 9$, $x - y = 7$ are examples of equation.

Teacher's Note

Teacher should bring real common balance in class or by any available material make a balance and explain equations by putting different objects or number cards in it.

9.2 LINEAR EQUATIONS

Define linear equation in one variable

An equation in which there is only one variable with degree one, is called linear equation in one variable.

Construct linear expression and linear equation in one variable

We know that an algebraic expression is a combination of numbers, variables and arithmetical operations.

Few examples of expression:

- (i) Four is added to twice of a number i.e. twice of a number plus four is: $2x + 4$
- (ii) Five is subtracted from thrice of a number i.e. thrice of a number minus five is: $3y - 5$.
- (iii) 6 is multiplied by a number i.e. six times of a number is: $6z$
- (iv) A number is divided by 8 i.e. eight part of a number is: $\frac{m}{8}$
- (v) Two is added to thrice of a number i.e. thrice of a number plus two is: $3n + 2$
- (vi) Five times a number is divided by 7 i.e. seventh part of five times of number is: $\frac{5a}{7}$

All the above are algebraic expressions in one variable.

We also know that an algebraic equation is a statement involving the sign of equality. Let us learn how to construct a linear equation.

- (i) Four is added to twice of a number the result is 10
i.e. $2x + 4 = 10$
- (ii) Three subtracted from thrice of a number the result is 1
i.e. $3y - 3 = 1$
- (iii) Three multiplied by a number is 9 i.e. $3z = 9$
- (iv) A number divided by 8 is 3 i.e. $\frac{m}{8} = 3$
- (v) Two added to thrice of a number is 5 i.e. $3n + 2 = 5$
- (vi) Five times of a number divided by 7 is 10 i.e. $\frac{5a}{7} = 10$

All these are linear equations in one variable.

Example 1: Construct the linear equations in one variable of the following.

- (i) Four times a number increased by 5 equals 11.
- (ii) If a number is subtracted from another number which is 5 more than thrice the first number, the result is 31.

Solution:

- (i) Let x be the number. Then four times of x is $4x$ and increasing this by 5 gives $4x + 5$ which is equal to 11. Hence the required equation is:

$$4x + 5 = 11$$

- (ii) Let x denote the number.
Then thrice the number is $3x$. Now other number which is 5 more than $3x$ is $3x + 5$. The difference of $3x + 5$ and x is $(3x + 5) - x$ which is equal to 31.

Hence the required equation is:

$$(3x + 5) - x = 31$$

EXERCISE 9.1

Construct the following linear equations.

- (1) The difference of a number and twenty is thirty two.
- (2) The sum of a number and 8 is 15.
- (3) Five multiplied by a number is 35.
- (4) Sixteen divided by four is equal to 4.
- (5) A number increased by two is 4.
- (6) Fifteen decreased from a number, the result is 5.
- (7) Twice of a number increased by 4 is 20.
- (8) My age is 15 years greater than my brother's age.

Solve simple linear equations involving fractional and decimal coefficient like $\frac{1}{2}x + 5 = x - \frac{1}{3}$.

Consider a problem in which there is weight of 10 gm on one pan of physical balance and there are two weights on other pan one is 8gm and other is unknown.

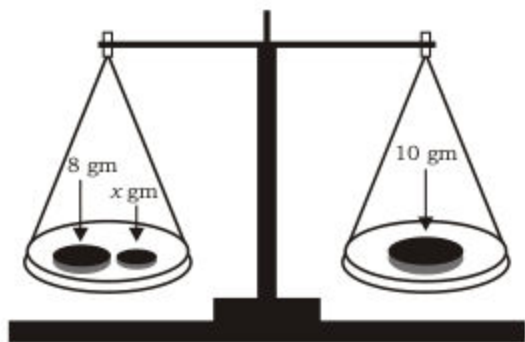
Let us represent the unknown weight by x . So this problem is written as: $x + 8 = 10$ to keep the balance.

If we put $x = 2$, then $2 + 8 = 10$ is true statement.

Thus, a weight of 2 grams is placed with the weight of 8 grams to keep the scale in balance.

Here the sentence $x + 8 = 10$ is called equation and 2 is its solution.

This equation contains one variable and its degree is also one. Such equation is called **linear equation** in one variable as we have already studied.



Following are some examples of linear equations.

$$x = 4, \quad x + 8 = 15, \quad 5y = 30 \\ \text{and } 5 - 2 = 3t + 8$$

An open sentence which contains the symbol “=” is called equations. The value of the variable which makes the equation true, is called the solution of the equation.



Activity

Let us solve $x + 5 = 8$. Putting different values of x , we get.

For $x = 1$, we get $1 + 5 = 8$, which is a false statement.

For $x = 2$, we get $2 + 5 = 8$, which is a false statement.

For $x = 3$, we get $3 + 5 = 8$, which is a true statement.

Construct any two linear equation in one variable.

Example 1: Solve $x + 5 = 11$

Solution:

$$x + 5 = 11$$

Subtracting 5 from both sides, we get.

$$x + 5 - 5 = 11 - 5$$

or $x + 0 = 6$

or $x = 6$

Same number can be subtracted from both sides of the equation.

Example 2: Solve $x - 8 = 21$

Solution: $x - 8 = 21$

Adding 8 to both sides we get

$$x - 8 + 8 = 21 + 8$$

or $x = 29$

checking $x - 8 = 21$

$$29 - 8 = 21 \quad (\text{by putting } x = 29)$$

or $21 = 21$ which is a true sentence.

Same number can be added to the both sides of the equation.

Example 3: Solve: $7x = 84$

Solution: $7x = 84$

Dividing both sides by 7, we get

$$\frac{7x}{7} = \frac{84}{7}$$

or $x = 12$

checking $7x = 84$

$$7 \times 12 = 84 \quad (\text{by putting } x = 12)$$

or $84 = 84$ which is a true sentence.

Both sides of equation can be divided by the same non-zero number.

Example 4: Solve $\frac{x}{3} = 7$

Solution: $\frac{x}{3} = 7$

Multiplying both sides by 3, we get

$$\frac{x}{3} \times 3 = 7 \times 3$$

or $x = 21$

checking: $\frac{x}{3} = 7$

$$\frac{21}{3} = 7 \quad (\text{by putting } x = 21)$$

or $7 = 7$ which is a true sentence

Both sides of equation can be multiplied by the same number.

Both sides of the equation can be interchanged

Let us solve some more equations.

Example 5: Solve: $3t + 5 = 20$

Solution: $3t + 5 = 20$

$$3t + 5 - 5 = 20 - 5 \quad (\text{subtracting 5 from both sides})$$

or $3t = 15$

or $\frac{3t}{3} = \frac{15}{3}$ (Dividing both sides by 3)

or $t = 5$

Verification

$$3t + 5 = 20$$

By using $t = 5$

we get $3(5) + 5 = 20$

or $15 + 5 = 20$

or $20 = 20$

Which is true sentence hence verified.

Example 6: Find the solution of the following equations:

(i) $\frac{1}{2}a + 5 = a - \frac{1}{3}$ (ii) $\frac{x+1}{2} = \frac{x+4}{3}$ (iii) $\frac{3x+4}{5+2x} = 1$

Solution: (i) $\frac{1}{2}a + 5 = a - \frac{1}{3}$

multiplying both sides by L.C.M 6.

$$6\left(\frac{1}{2}a + 5\right) = 6\left(a - \frac{1}{3}\right)$$

$$\cancel{6}\left(\frac{1}{\cancel{2}}a\right) + 6(5) = 6(a) - \cancel{6}\left(\frac{1}{\cancel{3}}\right)$$

$$3a + 30 = 6a - 2$$

$$30 + 2 = 6a - 3a$$

$$32 = 3a$$

$$\text{or } 3a = 32$$

$$a = \frac{32}{3}$$

Solution: (ii) $\frac{x+1}{2} = \frac{x+4}{3}$

$$6 \times \left(\frac{x+1}{2}\right) = 6 \times \left(\frac{x+4}{3}\right) \quad (x+4) \text{ (Multiplying by LCM 6)}$$

$$3(x+1) = 2(x+4)$$

$$3x + 3 = 2x + 8$$

$$3x - 2x = 8 - 3$$

$$x = 5$$

Solution: (iii) $\frac{3x+4}{5+2x} = 1$

$$(\cancel{5+2x}) \times \left(\frac{3x+4}{\cancel{5+2x}}\right) = (5+2x) \times 1 \quad \text{(Multiplying both sides (5+2x))}$$

$$3x + 4 = 1(5 + 2x)$$

$$3x + 4 = 5 + 2x$$

$$3x - 2x = 5 - 4$$

$$x = 1$$

EXERCISE 9.2

Solve the following equations.

(1) $x - 7 = 8$

(2) $y + 5 = 12$

(3) $x + 2 = 8$

(4) $7y = 14$

(5) $3x = 30$

(6) $y - 16 = 2$

(7) $\frac{x}{4} = 3$

(8) $6x + 1 = 49$

(9) $8x = 32$

(10) $7y + 2 = 44$

(11) $6 = x + 9$

(12) $9x - 21 = 78$

(13) $\frac{6x-4}{2x+2} = 2$

(14) $\frac{x}{2} + 4 = x - \frac{1}{3}$

(15) $\frac{5x-4}{8} = \frac{x+6}{4}$

Solve real life problems involving linear equations

The method of solving word problems is explained through the following examples.

To solve a word problem remember the following steps.

- Read the problem carefully. Draw its outline and
- Find what is asked.
- Whatever is asked represent it by a variable x or y .
- Construct an equation according to the given conditions.
- Solve the equation to get required answer.
- Check the answer so obtained.

Example 1: Hashim and Qasim both deposited a sum of Rs 47 in the fund of Hilal-e-Ahmer. If the Hashim's share was Rs 25. Find the share of Qasim.

Solution: To solve this problem first read it carefully. In this problem share of Qasim is to be found.

Deposited by both	= 47 rupees
Suppose the share of Qasim	= x rupees
Share of Hashim	= 25 rupees
Share of both	= $x + 25$

Teacher's Note

Teacher should give more real life examples for practice to students to clear their concepts more effectively about linear equations.

So, according to the given condition of the problem, we have

$$x + 25 = 47$$

Let us solve it

$$x + 25 = 47$$

$$\text{or} \quad x + 25 - 25 = 47 - 25$$

$$\text{or} \quad x = 22$$

Hence share of Qasim is Rs 22.

Example 2: Zubeda and Farah jointly planned to go to visit Zoo for entertainment. They both collected Rs 44. Farah paid Rs 6 more than Zubeda. Find the share of Zubeda.

Solution:

Suppose Zubeda paid = x rupees

Farah Paid (Rs 6 more) = $x + 6$ rupees

Total amount = $x + (x + 6)$ rupees

Amount collected = 44 rupees

So, according to the condition of the problem

$$x + (x + 6) = 44$$

$$x + (x + 6) = 44$$

$$\text{or} \quad x + x + 6 = 44$$

$$\text{or} \quad 2x + 6 = 44$$

$$\text{or} \quad 2x + 6 - 6 = 44 - 6$$

$$\text{or} \quad 2x = 38$$

$$\text{or} \quad \frac{2x}{2} = \frac{38}{2}$$

$$\text{or} \quad x = 19$$

Hence, Zubeda paid Rs 19

Example 3: If the sum of a number and 8 is multiplied by 5, then 60 is obtained. Find the number.

Solution: Let the required number be x .

$$\text{Sum of number and 8} = x + 8$$

$$\text{Sum multiplied by 5} = 5(x + 8)$$

$$\text{Given product of sum and 5} = 60$$

So, according to the condition of the problem

$$5(x + 8) = 60$$

$$\text{or} \quad 5x + 40 = 60$$

$$\text{or} \quad 5x + 40 - 40 = 60 - 40$$

$$\text{or} \quad 5x = 20$$

$$\text{or} \quad \frac{5x}{5} = \frac{20}{5}$$

$$\text{or} \quad x = 4$$

Hence, the required number is 4.

EXERCISE 9.3

1. Sara stitches 44 mirror pieces on a red and green shirt. If 6 more mirror pieces are stitched on red shirt than green shirts, find the mirrors stitched on green shirt.
2. In a cricket match Jamil and Saleem enhanced the score of the team by 84 runs. If Jamil scored 12 runs more than Saleem. Find the score of the Saleem.
3. Sum of a number and its double is 9. Find the number.
4. The product of 7 and the sum of a number and 6 is 77. Find the number.
5. The sum of 6 and four times a number is 42. Find the number.

6. The sum of the ages of mother and her daughter is 22 years. The daughter is 20 years younger than mother. Find the age of her daughter.
7. When Dua opens a book there are two pages in front of her. The sum of the page numbers is 155. If one page number is 50, what is the other page number?
8. The price of a toy was decreased by Rs 7. If new price is 25 rupees. Find the original price.

REVIEW EXERCISE 9**1. Write the algebraic expression of the following:**

- (i) Sum of x and 4
- (ii) A number less than 7
- (iii) 9 times a number x
- (iv) A number y divided by 6
- (v) Sum of two numbers x and y
- (vi) Product of a variable x with itself

2. Write an equation for each of the following:

- (i) A number x decreased by 3 is 10
- (ii) Thrice a number y increased by 5 is 17
- (iii) Twice a number z more 4 is 20
- (iv) Half of a number m subtracted from 7 is 4
- (v) Sum of two consecutive even numbers is 16
- (vi) After 8 years I will be thrice as old as I am now

3. Solve the following equations:

- (i) $5x - 3 = 18$
- (ii) $\frac{y+1}{2} = 5$
- (iii) $\frac{m-1}{3} = 1$
- (iv) $4y - 10 = 1$
- (v) $\frac{p}{7} - 3 = 2$
- (vi) $\frac{1}{2}q = 3$

4. A pole x metres long is broken into two parts, one part is 3 metres long and other part is $2x - 17$ metres long. Find the original length of the pole.
5. If a number is tripled and the result is increased by 5, we get 44. Find the number.
6. Twice a number added to half of itself equals 20. Find the number.
7. Thrice a number decreased by 5 exceeds the number by a unit. Find the number.
8. The sum of two consecutive even integers is 34. Find the smallest of them.

SUMMARY

- An algebraic sentence involving the sign of equality “=” is called an **algebraic equation**.
- An algebraic equation in which there is only one variable and degree one is called **Linear Equation in One Variable** or **Simple Linear Equation**.
- ‘Solving a **Simple Linear Equation**’ means to find that value of **unknown variable** which satisfies the given equation.
- Several real life problems are, first of all, **converted into mathematical form** i.e linear equations and then solved for the unknown by using addition, subtraction, multiplication and division rules.